SESSION 17

AIOps for Enterprise

Enterprise AIOps Use Cases

- Incident Management Automation
- Customer Experience Monitoring
- Cloud Cost Optimization

Session 17 — Agenda

01

Incident Management Automation

Alert correlation, auto-remediation, runbook execution

02

Customer Experience Monitoring

Synthetic monitoring, RUM, NPS correlation with IT metrics

03

Cloud Cost Optimization

Anomaly detection, rightsizing, FinOps integration

04

Cross-cutting Patterns

MTTD/MTTR reduction, ROI metrics, governance

01

Incident Management Automation

Detect → Correlate → Remediate → Learn

What is Incident Management Automation?

AIOps-powered incident management uses ML and analytics to automate the end-to-end incident lifecycle — from detection through resolution — reducing human toil and accelerating recovery.

1

Detect

Real-time anomaly detection across logs, metrics & traces

→

2

Correlate

Group related alerts; eliminate noise with topology maps

→

3

Diagnose

Root-cause analysis with dependency graphs & ML models

→

4

Remediate

Trigger runbooks, self-healing scripts & auto-rollbacks

Alert Correlation & Noise Reduction

99%

Alert noise reduction

70%

Fewer tickets created

3×

Faster MTTD

Core Correlation Techniques

- Topology-aware grouping using CMDB/service maps
- Time-window clustering (sliding window algorithms)
- Semantic similarity via NLP on alert messages
- Temporal pattern mining for recurrent incidents
- Causal inference to distinguish root cause from symptoms

Auto-Remediation & Runbook Automation

Incident Detected

→

RCA Engine Activated

→

Runbook Selected via ML

→

Script / API Executed

→

Verification & Ticket Close

Key Benefits

MTTR reduced by up to 60%

Source: IBM (2024)

24/7 autonomous remediation

No human intervention required

Consistent runbook execution

Eliminates human error

Kubernetes & cloud-native support

Pod restart, scaling, rollback

Audit trail & compliance logging

SOX, HIPAA, ISO 27001

Real-World Case Study: Incident Automation

Goldman Sachs

Challenge:

~1M alerts/day; Ops team overwhelmed

Solution:

BigPanda AIOps for correlation & triage

Outcome:

90% alert reduction; 50% MTTR improvement

Deutsche Telekom

Challenge:

Network incidents impacting SLA compliance

Solution:

IBM Watson AIOps for RCA automation

Outcome:

60% fewer P1 incidents; auto-heal rate 42%

Shopify

Challenge:

High-traffic events causing cascading failures

Solution:

Custom AIOps + PagerDuty automation

Outcome:

Zero-touch resolution for 35% of incidents

02

Customer Experience Monitoring

Correlate IT health metrics with real user satisfaction

Customer Experience Monitoring with AIOps

AIOps bridges the gap between IT infrastructure health and business outcomes — connecting app performance data to real user satisfaction scores and revenue impact.

Synthetic Monitoring

Simulated user journeys run continuously from global PoPs. Catch regressions before real users are affected.

Tools: Catchpoint, Dynatrace, Selenium Grid

Real User Monitoring (RUM)

Capture actual user sessions: page load times, errors, clicks. Segment by geography, device, ISP.

Tools: New Relic Browser, Datadog RUM

NPS & CSAT Correlation

ML models correlate degraded response time with NPS drops. Quantify the $ value of performance improvements.

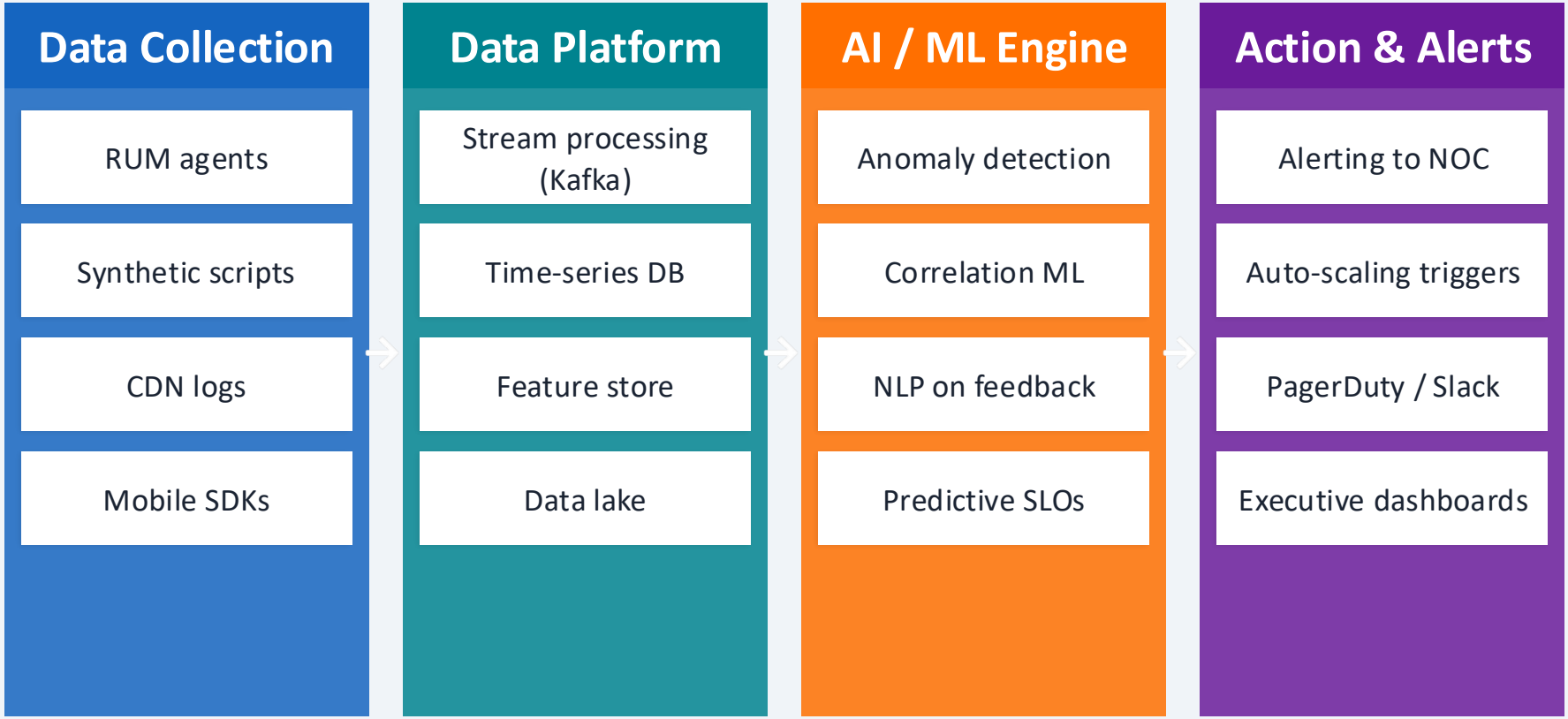
Tools: Medallia + APM integrations

RUM vs. Synthetic Monitoring — Key Differences

Attribute	Real User Monitoring (RUM)	Synthetic Monitoring
Data Source	Actual user browsers/apps	Scripted bots from PoPs
Coverage	100% of production traffic	Controlled test coverage
Proactive?	Reactive — issues detected after occurrence	Proactive — catches issues before users
Geo Insights	Real-world geography & ISP data	Simulated locations worldwide
Best For	Understanding real-world performance	Baseline & regression testing
Noise Level	Higher — real-world variability	Lower — controlled environment

💡 ***Best Practice: Use BOTH in tandem — synthetic for baselines, RUM for real-world insight.***

CX Monitoring Architecture



Use Case: E-Commerce CX Monitoring

1s

Page delay
= 7% conversion drop

53%

Mobile users leave after
3s load time

$2.5B

Amazon lost per year
to 1s slowdown (hypothetical)

How AIOps Helps E-Commerce

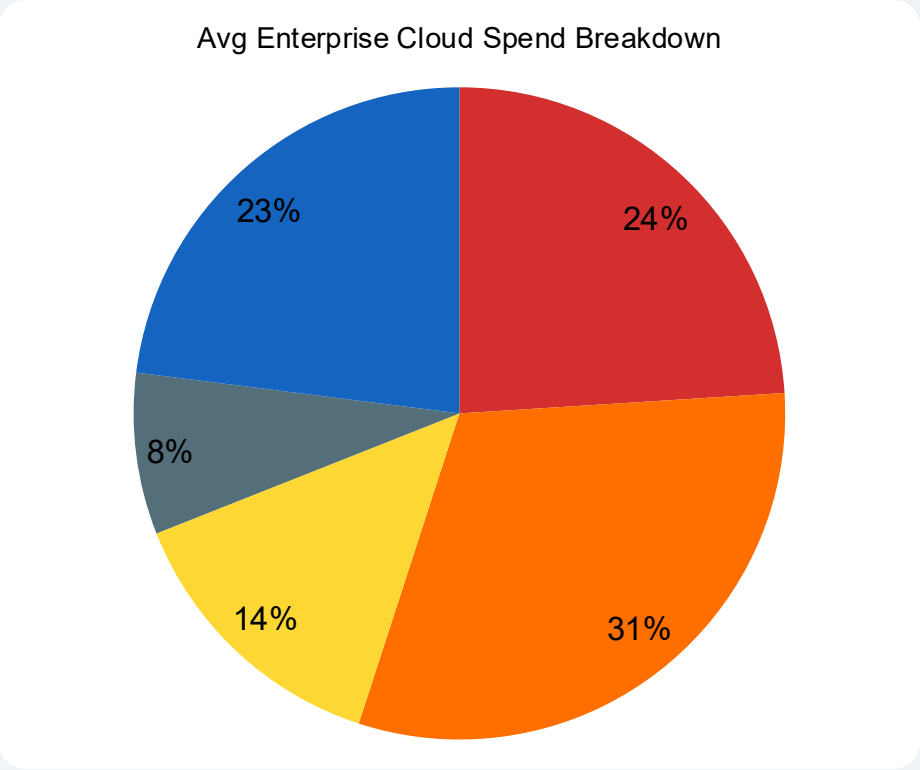
- Detect checkout funnel degradation before revenue impact
- Correlate CDN latency spikes with cart abandonment rate
- Auto-scale product catalog service on Black Friday traffic surges
- Flag third-party script failures (ads, payments) in real time
- Generate automated incident reports for revenue impact estimation

03

Cloud Cost Optimization with AIOps

Eliminate cloud waste · Right-size resources · Automate FinOps

The Cloud Waste Problem



Industry Stats

$147B

Wasted cloud spend globally in 2023

Flexera Cloud Report 2024

32%

Average cloud budget wasted per org

Gartner Cloud Optimization

68%

Orgs lack visibility into cloud costs

CloudZero Survey 2024

Ref: Flexera 2024 State of the Cloud Report | https://www.flexera.com/blog/cloud/cloud-computing-trends-state-of-the-cloud-report/

AIOps Capabilities for Cloud Cost Optimization

Anomaly Detection

Detect unexpected cost spikes in real time using statistical models (Z-score, Isolation Forest). Alert before month-end billing surprise.

Rightsizing Engine

Analyze CPU/memory utilization histograms. Recommend optimal instance type, size, and family using Bayesian optimization.

Scheduling Automation

ML-predicted load patterns drive automated start/stop of dev/test environments. Average savings: 30–40% on non-prod.

Reserved/Spot Advisor

Predict workload stability to recommend RI/SP commitments vs. Spot. Balance commitment vs. flexibility.

FinOps Dashboards

Unit economics views — cost per customer, per feature, per pipeline run. Chargeback & showback automation.

Forecast & Budget

LSTM-based cost forecasting with 90-day horizon. Automated budget alerts with confidence intervals.

Case Study: Cloud Cost Optimization with AIOps

Netflix

Before:

AWS bill growing 35% YoY; manual cost reviews quarterly

After:

AIOps anomaly detection + auto-rightsizing + spot optimization

Savings:

~$50M/yr saved; 99.99% availability maintained

Source:

Netflix Tech Blog 2023

Lyft

Before:

30% idle instances during off-peak; manual scheduling infeasible

After:

ML-based predictive autoscaling + automated shutdown policies

Savings:

25% EC2 cost reduction; MTTR improved by 40%

Source:

Lyft Engineering Blog 2023

Spotify

Before:

Multi-cloud cost visibility gap; eng teams overspending on GCP

After:

Harness CCM + AIOps anomaly alerts + chargeback dashboards

Savings:

$12M annual savings; 3× ROI in Year 1

Source:

CNCF KubeCon Talk 2023

Ref: Netflix Tech Blog | https://netflixtechblog.com | Lyft Engineering | https://eng.lyft.com | CNCF | https://www.cncf.io

Cross-Cutting: AIOps KPIs & Success Metrics

Incident Mgmt	CX Monitoring	Cloud Cost
MTTD (Mean Time to Detect)	Apdex Score	Cloud Cost per Unit of Work
MTTR (Mean Time to Resolve)	Core Web Vitals (LCP, FID, CLS)	Resource Utilization Rate (%)
Alert-to-Incident Ratio	Conversion Rate vs. Latency	Rightsizing Coverage (%)
Auto-resolve Rate (%)	NPS / CSAT Correlation	RI/SP Commitment Ratio

Ref: DORA Metrics | https://dora.dev | Google SRE Book | https://sre.google/sre-book/table-of-contents/

AIOps Tools & Platforms Landscape

Incident Management	CX Monitoring	Cloud Cost
PagerDuty AIOps	Dynatrace	AWS Cost Anomaly
BigPanda	New Relic	Harness CCM
Moogsoft	Datadog RUM	CloudHealth
IBM Watson AIOps	Catchpoint	Apptio Cloudability
Splunk ITSI	Elastic Observability	Spot.io

Ref: Gartner Magic Quadrant for AIOps Platforms 2024 | https://www.gartner.com/en/documents/aiops-platforms

Enterprise AIOps Implementation Roadmap

Phase 1 Foundation

0–3 months

- Instrument with observability agents
- Build centralized data lake
- Define SLOs & baselines
- Tag cloud resources

Phase 2 Detection

3–6 months

- Deploy anomaly detection models
- Implement alert correlation
- Launch RUM & synthetic monitoring
- Cloud cost dashboards live

Phase 3 Automation

6–12 months

- Runbook automation for top-10 incidents
- Auto-remediation for self-healing
- Rightsizing automation pipelines
- CX / IT metric correlation

Phase 4 Optimize

12+ months

- Continuous ML model retraining
- Predictive incident prevention
- FinOps governance policies
- Executive ROI dashboards

Challenges & Best Practices

Common Challenges	Best Practices
Data silos across tools & teams	Start with highest-impact, low-complexity use cases
Alert fatigue from poorly tuned models	Invest in data quality before ML models
Lack of contextual topology maps	Build a full-stack service topology/CMDB first
Resistance from Ops teams (cultural change)	Run AIOps in shadow mode before automation
Model drift in dynamic cloud environments	Continuously retrain models with feedback loops

Session 17 — Key Takeaways

01

Incident Automation:

AIOps reduces MTTD/MTTR by up to 60%, auto-correlates noisy alerts, and executes runbooks without human intervention.

02

CX Monitoring:

Synthetic + RUM monitoring, correlated with NPS/CSAT, enables proactive protection of digital customer journeys.

03

Cloud Cost Optimization:

AIOps-powered anomaly detection, rightsizing engines, and FinOps dashboards can cut cloud waste by 25–40%.

04

Cross-Cutting Theme:

Success requires: data quality, service topology, iterative automation, and continuous ML retraining.